

Multiple Choice (10 marks)

1. Primary motor cortex activity results in
 - (a) bilateral contraction of antigravity limb muscles.
 - (b) bilateral contraction of limb musculature.
 - (c) contraction of ipsilateral limb musculature.
 - (d) contraction of contralateral limb musculature.
2. When asked where they would like to live for the rest of their lives most older adults say that they would prefer
 - (a) Moving to be close to their children
 - (b) Moving to Arizona or Florida if they could
 - (c) Residing in a very nice nursing home
 - (d) Staying in their own home
3. Which is a major characteristic of the bunyavirus family?
 - (a) Amplify by persistent infection of midges, mosquitoes, sand flies, ticks
 - (b) Predominantly mosquito borne
 - (c) Genetic recombination or reassortment
 - (d) Spread to humans via bites of mice and rats
4. Repeat core sequences consisting of 2, 3, or 4 base pairs are known as what?
 - (a) Single nucleotide polymorphisms (SNPs)
 - (b) Microsatellites
 - (c) Minisatellites
 - (d) Satellites
5. The stage of meiosis in which chromosomes pair and cross over is:
 - (a) prophase I
 - (b) metaphase I
 - (c) prophase II
 - (d) metaphase II

True / False (10 marks)

Answer True or False

6. True or False: Monosomy is the most frequently observed chromosome abnormality in first trimester spontaneous miscarriages.
7. True or False: Girls most frequently learn about masturbation through self-discovery rather than external sources.
8. True or False: Cytotoxic T cells are activated by reacting with antibodies that bind to viral particles in the bloodstream.
9. True or False: Herpes viruses do not cause infantile paralysis.
10. True or False: Sterilization is the most common method of birth control for married couples in the U.S.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of proprioceptive information in the function of the temporomandibular joint, specifically focusing on the contributions of the masseteric and auriculotemporal nerves.
12. Discuss the process of ovulation within a typical 28-day menstrual cycle, focusing on the timing, physiological changes, and factors influencing ovulation.

End of Paper